Channel-Aware Training, Not Closed-Loop Control, Drives Robust Regeneration in Neural Cellular Automata

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Abstract

We set out to test whether closed-loop control of global chemical signals makes regenerative Neural Cellular Automata (NCAs) more robust, and built the evaluation to prove it: recurring multi-block lesions larger than the perception radius, a small evolved controller for three broadcast modulator channels, and held-out damage seeds. On a single trained model, the experiment looked like a success. However, crossing five independently trained parent seeds (two independent controller evolutions each; preregistered) dissolves that success into three findings. First, the robustness we had attributed to control is an effect of training, not control — but only when the channels actually carry signal: the study's parents (channels present yet silent throughout training) beat unmodulated siblings in five of five seeds under the preregistered metric, yet post-hoc controls (own-grown initial states, translation-aligned scoring) reduce that to a single-seed effect, while fresh preregistered retrains with random modulator inputs beat paired $K{=}0$ siblings in five of five seeds *under the corrected protocol*. Growing up under modulation — even random modulation — is what confers robustness; the original comparison never exercised the channels it credited. Second, evolved controllers add little on top: on four parents the effect is absent or noise-level; on the fragile parent both evolutions reproducibly help (0.035–$0.038 \rightarrow 0.029$–0.030, survival $0.975 \rightarrow 1.00$); yet its tonic vector is nearly identical (cosine 0.99) to a sibling's that does not help, so the benefit is an interaction with parent dynamics, not a distinctive operating point. All ten controllers emit flat, nonzero, parent-specific constants with no lesion-locked response. Third, these constants are organism-locked: a single-donor probe penalizes five of five siblings (lethally in two), a full $5{\times}5$ transfer matrix shows most transfers harmful and none better than the recipient's own controller, and injecting each donor's tonic constant directly, no controller at all, reproduces its transfer outcomes in 46 of 50 cells, including all lethal ones. Single-parent evaluation would have approved all of it, making the attribution protocol itself the transferable artifact.

1 Introduction

Growing Neural Cellular Automata (GNCAs) grow a target morphology from a single seed and regenerate it after damage using one local, differentiable update rule [1]. The catch is perception: a cell sees only its immediate neighborhood, so wounds larger than a few cells contain tissue with no signal to integrate, and a severed fragment has no way to know what it was part of. The natural remedy is a global signal. Prior work, our own single-parent study included (Appendix H), adds global modulator channels and evolves release policies for them, reporting the outcome on one trained model.

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That single-model habit is the problem this paper is about. When a modulated NCA beats an unmodulated one, the channels being *present during training*, the controller *acting at run time*, and whatever *transfers* to another organism are confounded. Our single-parent study learned this the hard way: it credited closed-loop control for a $2\times$ improvement that vanished under cross-parent testing, where the controller transferred lethally to siblings while zero-modulation channel parents beat the unmodulated baseline in every seed (Appendix C). The improvement was real, but the attribution was wrong.

We therefore ran the preregistered attribution study that the original design needed with primary statistic and decision thresholds fixed before data collection. Five parent seeds, each trained from scratch as a $K{=}0$ pair and a channel-aware $K{=}3$ pair; *two* independent controller evolutions per channel parent; four conditions per seed, all on held-out damage seeds; then, post-hoc, a full $5{\times}5$ transfer matrix and a tonic-transplant condition (Section 3; preregistrations and analysis scripts timestamped before the runs).

Contributions.

1. A preregistered robustness-attribution protocol for regenerative NCAs: adversarial damage calibrated to exceed perception, crossed with parent-seed variation, decomposed into E_{train}, E_{ctrl}, E_{transfer}, with calibration probes that keep the evolution landscape interpretable (Appendix F).
2. The attribution, settled causally: training with channels is the robust cause (5/5 seeds); the controller effect is parent- dependent (absent or noise-level on four parents, reproducible on the fragile one); the evolved artifact is a parent-specific tonic constant that penalizes every sibling it meets (lethally in two), and injecting the constant alone reproduces its transfer outcomes: the matrix (23/40 harmful, none beating the recipient's own) closes the case.

2 Related Work

Neural cellular automata. GNCAs grow patterns from a seed and regenerate after damage [1], with extensions to self-classification [2] and textures [3]. All rely on local perception, and regeneration degrades beyond the perception radius; the regime that our benchmark exploits. Signal channels [4], goal conditioning [5], and information-dynamical analyses of self-maintenance [6] all treat the signal as fixed input or emergent byproduct instead of a controlled output; we ask what is actually helping when such a channel *appears* to help.

Search over evaluation regimes. Our design follows a concern shared by novelty search [7] and quality-diversity optimization [8]: a fixed objective can misdirect search and hide failure modes. Environment-generation methods co-evolve challenges with solutions. POET [9], PAIRED [10], ACCEL [11]; Our benchmark does not yet evolve the damage distribution and serves as the stationary baseline such co-evolution can be judged against.

3 Experimental design

Damage regime. Recurring multi-block lesions; every 150 steps, $n{=}4$ contiguous $16{\times}16$ blocks at seeded positions, $T{=}2000$ (the information-isolation regime of Appendix E). Damage seeds 0–7 drive evolution; held-out seeds 10000–10007 drive all reported numbers; the schedule is deliberately adversarial (Appendix F).

Parents and controllers. Per seed s we train a $K{=}0$ and a channel-aware $K{=}3$ parent from scratch; "channel-aware" means the three modulator inputs are part of the architecture during training, with their injected level pinned to the neutral vector throughout parent training (Appendix D); the $K{=}3$ parent's three global modulator channels are the only non-local pathway, each carrying a *tonic* (slow baseline) and *phasic* (event-triggered, fast-decaying) component: the distinction from biological neuromodulation. A 259-parameter controller ($4{\to}32{\to}3$, $\tanh$) reads four target-free grid statistics and sets release every 10 steps; each parent gets *two* independently evolved controllers via CMA-ES [13] in Evosax [14] (population 64, $\sigma_0{=}0.01$, 300 generations, different evolution seeds, event-weighted Hamming objective; Appendix G).

Metrics. Final Hamming is the mismatch rate, at the final step, between the binarized alive mask ($\alpha > 0.5$) and the target's binarized α mask. Survival is the fraction of runs (damage seed $\times$ condition seed) whose final Hamming falls below 0.1; a condition is *lethal* when survival is 0.00, i.e., every run ends above the threshold. Hamming is computed against a fixed target frame and so conflates morphological error with whole-organism translation drift (Appendix H); a translation-aligned variant is reported alongside a new initial-state control in Appendix I.

Preregistered comparison. All conditions run on the same held-out damage seeds (5 condition seeds $\times$ 8 damage seeds) from a shared $t{=}0$ state grown by the $K{=}3$ parent. The shared start is itself a loaded choice: the 12 non-RGBA channels of a grown state are a network-specific latent code, so the $K{=}0$ condition begins from an out-of-distribution initial state — a $K{=}3$-grown morphology living under $K{=}0$ dynamics. Holding the starting morphology constant isolates the update rule, but it disadvantages the $K{=}0$ condition, and the most fragile pair (s1: survival 0.05–0.08 vs. 0.975) is exactly where this could bite; a post-hoc control re-evaluating every $K{=}0$ parent from its own-grown state is reported in Appendix I. The primary statistic is $\Delta_s = H(K{=}3, m{=}0, s) - H(K{=}3, \text{own ctrl}, s)$, positive when the evolved controller helps *its own* parent. Fixed before data collection: *controller effect supported* if $\Delta_s > 0$ in $\geq 4/5$ seeds with sign-consistent differences; *channel-training effect supported* if $H(K0, s) - H(m0, s) > 0$ in $\geq 4/5$; *transfer failure supported* if the pilot donor controller underperforms the own-controller substantially (preregistered criterion; operationalized as > 0.005 Hamming or survival < 0.9, §4) in $\geq 4/5$. Effects are reported per seed (median and range); no significance claims at five seeds. The defense study reported here fixed the two-evolutions-per-parent rule in its preregistration before launch (`experiment_results/20260818_evoseed_defense/PREREGISTRATION.md` in the repository history). Controller-output (m_t) series distinguish tonic calibration from event-locked policy (Appendix B). A post-hoc follow-up evaluates the full $5{\times}5$ transfer matrix among the defense parents (both controller replicas, 8 damage $\times$ 3 condition seeds per cell; Table A3) plus a *tonic transplant* injecting each donor's realized mean m_t as a constant.

4 Results

Table 1 reports the preregistered decomposition; the full per-seed, per-condition numbers behind it are in Appendix A.

Table 1: Final Hamming (mean over 5 condition seeds $\times$ 8 held-out damage seeds). Survival in parentheses where it departs from 1.00. Architecture-present training beats $K{=}0$ in 5/5 seeds — attribution entangled (silent training channels, Appendix D; shared-state protocol, §3; post-hoc controls, Appendix I); the controller effect is parent-dependent.

Parent seed	$K{=}0$	$K{=}3$, $m{=}0$	Own ctrl (2 runs)	Interpretation
0	0.035–0.036	0.029	0.029–0.030	channel effect; no control effect
1	0.163–0.175 (surv. 0.05–0.08)	0.035–0.038 (surv. 0.975)	0.029–0.030 (surv. 1.00)	channel effect + replicated tonic benefit
2	0.030	0.018–0.021	0.022–0.026	channel effect; controller slightly worse
3	0.049–0.052	0.028–0.029	0.032–0.033	channel effect; controller slightly worse
4	0.042–0.045	0.033	0.034–0.035	channel effect; no control effect

Outcome. *Channel-training supported* (positive in 5/5 seeds; bar $\geq$4/5); *transfer failure supported* (penalty 5/5, lethal in 2). Under the preregistered rule the controller effect is classified as *parent-dependent*, preregistered **Outcome C**: noise-level on four parents, reproducibly beneficial on the fragile one (both evolutions: 0.035–$0.038 \rightarrow 0.029$–0.030, survival $0.975 \rightarrow 1.00$). The channel-training criterion was met as preregistered, but the comparison it names is coarser than the phrase suggests: the channels were silent throughout parent training, so the effect cannot be modulatory experience (Appendix D), and the $K{=}0$ sibling starts from the $K{=}3$-grown state (§3). Appendix I separates what these confounds leave separable: the 5/5 reduces to protocol and drift effects on healthy parents plus one genuinely fragile parent (s1); preregistered retrains with exercised channels (Appendix J) restore a genuine $5/5$ training effect on the corrected protocol.

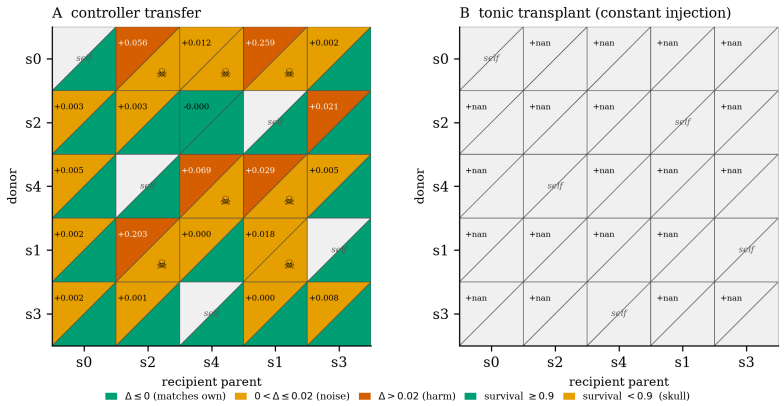


Figure 1: Transfer redesigned for attribution (replica-averaged; per-cell Δ from the recipient's own controller; survival as the split lower triangle; rows/columns cosine-ordered so the aligned pair s4–s1 borders the diagonal and the anti-aligned pair s3–s2 sits at opposite ends). Panel A: controllers. Panel B: the same matrix with each donor's tonic constant injected instead, matching A in 46/50 cells, every lethal one included. Values: Table A3.

What the controller actually emits. All ten evolved controllers emit a *constant at a nonzero level*: within-rollout per-channel std of m_t is 0.0002–0.0046, tonic with no lesion-locked response in 10/10 runs, while channel means sit at parent-distinct offsets (Figure A1). Evolution neither collapses to neutral output nor discovers a policy: it finds a parent-specific tonic calibration (whose predictive limits are discussed in Section 5).

Cross-parent transfer matrix. The full 5×5 matrix among the defense parents, both replicas, three condition seeds per cell, 40 off-diagonal cells (Table A3, Figure 1), 23/40 transfers are harmful (survival < 0.9 or final Hamming worse than the recipient's zero-output baseline by > 0.005), 15 are indistinguishable from zero-output, and 2 beat it. No foreign controller beats the recipient's own controller beyond noise. Lethal transfer replicates across controller replicas and condition seeds: donor s3 kills recipient s2 in both replicas (survival 0.00); donor s0 on recipient s1 is lethal in e1 and near-lethal in e2 (Figure 1A).

Tonic transplant. Injecting each donor's realized mean m_t as a constant in place of its controller (both replicas, three condition seeds) reproduces the full controller's outcome in 36/40 off-diagonal cells, including every lethal transfer (the four mismatches are magnitude differences inside already-lethal cells), and matches the recipient's own controller in all 10 diagonal cases (46/50 overall; Appendix A). The transferable component is thus the tonic setpoint; the controllers' small dynamic residual did not measurably contribute.

5 Discussion

Which component causes robustness. The trained organism, not run-time control. Parents trained with the channel architecture present beat their unmodulated siblings on every seed, even with modulation pinned neutral at evaluation. We initially read this as a developmental scaffold — the organism more robust because it grew up under global signals. While preparing the camera-ready revision we audited the training configuration and found the channel inputs pinned to the neutral vector for the whole of parent training: nothing modulatory was ever experienced, the channel weights sit at their initialization, and the $m{=}0$ parent is functionally a $K{=}0$ network with a different initialization draw (Appendix D). The 5/5 direction does not survive the camera-ready controls: from each parent's own-grown initial state, s4 flips on the plain metric (0.029 vs. 0.033), and under translation-aligned scoring all four healthy-parent comparisons sit within noise and nominally favor $K{=}0$ — only the fragile s1 pair retains a clear gap (aligned 0.026 vs. 0.071), a substantially intrinsic fragility of that parent draw that the shared-state protocol amplified (survival $0.10 \rightarrow 0.50$ from its own state; Appendix I). Preregistered retrains close the loop (Appendix J): parents trained

with random modulator inputs beat fresh paired $K{=}0$ siblings in $5/5$ seeds on the corrected protocol (median gap $+0.012$), while fresh silent-channel pairs show no such effect — restoring the scaffold reading in its precise form: *channel exercise during training*, not architecture or initialization luck, is what transfers robustness. What evolution adds is narrow: nothing measurable on four parents; on the fragile parent, a final stabilization that both independent evolutions find. Since that parent's tonic vector is nearly identical to a sibling's that gains nothing from it, the benefit is not a property of the tonic.

Why cross-parent transfer fails, and when it does not. The full matrix (Table A3) sharpens the single-donor probe: no foreign controller beats the recipient's own controller, most transfers are harmful (23/40), and the only beneficial ones stay within the tonic-aligned pair (s4→s1). The tonic transplant makes it causal: what transfers, for good or ill, *is* the constant (36/40 off-diagonal, 10/10 diagonal, every lethal cell included). Each controller's output is a calibration against its own parent's channel weights, so the same release level drives different dynamics in a sibling; evolution found an operating point for one organism, not a modulation law.

Limitations and future work. Five parents, two evolutions each; one morphology; one damage family; $n{=}5$ puts a one-sided sign test at $p{\approx}0.03$ for the 5/5 training condition — suggestive, not separable from initialization luck at this sample size. Only one parent in five proved controller-responsive, and the null controller result may partly reflect the search setup rather than the organism's capacity: the damage process is stationary and memoryless, a regime in which a fixed tonic is structurally optimal (Appendix H), and CMA-ES ran 300 generations from a near-neutral start. Two observations argue against reading the null as pure search failure: both independent evolutions converged to the same tonic family on every parent, and the pathway is high-gain (random actuation is lethal; constant levels alone swing mean Hamming from 0.003 to 0.93), so exploitable signal was available. The random-input retrains and added initialization pairs are now run (Appendix J); the open steps are co-trained parent–controller loops and non-stationary, diversity-driven damage, for which this benchmark is the controlled baseline.

Conclusion. In regenerative NCAs under adversarial recurring damage, robustness is built in development, not by the run-time controller: the study's silent-channel comparison dissolves under corrected controls, while fresh preregistered retrains — channels exercised by random inputs — restore a genuine $5/5$ advantage: growing up under modulation, even random modulation, is what confers robustness. Evolved modulation remains a parent-dependent tonic calibration, noise-level on most parents, beneficial on the fragile one, parent-locked under transfer. Attribute robustness across model seeds, not only damage seeds — and across initial-state provenance.

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A Full per-seed, per-condition results

Tonic transplant. Replacing each donor controller by its realized mean m_t injected as a constant (same protocol, both replicas, 3 condition seeds) reproduces the controller's outcome in 36/40 off-diagonal cells and 10/10 diagonal cells (46/50 overall; Figure 1B), including every lethal cell, where the transplant kills exactly as the controller does. The four mismatches are magnitude differences inside already-lethal cells. Under the tested regime, the controllers' small dynamic residual did not measurably contribute, on-parent or off (full CSVs in the repository).

AUC mirrors the same ordering in every seed (full CSVs in the repository).

B Controller-output (m_t) diagnostics

The tonic vector alone does not predict benefit: seeds 1 and 4 emit nearly identical tonic vectors (cosine similarity 0.993, Euclidean distance 0.0065), yet only seed 1 benefits from its controller. The

Table A1: Final Hamming (mean ± SD, 5 condition seeds × 8 held-out damage seeds) and survival for each of the ten runs (5 parent seeds × 2 independent controller evolutions) under hard recurring multi-block damage, T=2000. Conditions: K=3 parent with modulation pinned to zero; the parent's own evolved controller; the K=0 parent.

	K=3, m=0		K=3, own ctrl		K=0	
Run	H	surv	H	surv	H	surv
s0_e1	0.0290±.0135	1.000	0.0303±.0134	1.000	0.0361±.0137	1.000
s0_e2	0.0294±.0144	1.000	0.0292±.0136	1.000	0.0354±.0109	1.000
s1_e1	0.0384±.0266	0.975	0.0286±.0177	1.000	0.1628±.0389	0.075
s1_e2	0.0353±.0256	0.975	0.0301±.0210	1.000	0.1746±.0356	0.050
s2_e1	0.0183±.0145	1.000	0.0225±.0149	1.000	0.0299±.0178	1.000
s2_e2	0.0208±.0170	1.000	0.0258±.0161	1.000	0.0296±.0174	1.000
s3_e1	0.0281±.0177	1.000	0.0332±.0165	1.000	0.0491±.0084	1.000
s3_e2	0.0289±.0182	1.000	0.0321±.0164	1.000	0.0525±.0136	1.000
s4_e1	0.0332±.0233	1.000	0.0336±.0223	1.000	0.0420±.0111	1.000
s4_e2	0.0332±.0225	1.000	0.0345±.0222	1.000	0.0452±.0093	1.000

Table A2: Cross-parent transfer probe (unchanged from the per-parent study): the pilot donor controller evaluated on each recipient parent. Final Hamming and survival on held-out damage seeds; transfer is a penalty in 5/5 recipients, lethal in 2.

Recipient seed	Final Hamming	Survival
0	0.036	1.00
1	0.249	0.00 (lethal)
2	0.434	0.00 (lethal)
3	0.100	0.45
4	0.041	1.00

benefit arises from the interaction between a tonic calibration and parent-specific local dynamics, not from a distinctive chemical operating point.

Tonic alignment structures transfer. All lethal instances pair strongly negative tonic-vector cosines (s3→s2: -0.61; s0→s1: -0.93), while the only beneficial transfers, beating zero-output and matching the own controller, occur exclusively within the tonic-aligned pair s4→s1 (cosine $+0.99$, both replicas; adjacent to the diagonal in Figure 1). Across all 40 cells, tonic cosine correlates with transfer penalty ($r = -0.30$) and survival ($r = +0.34$); alignment is not sufficient, as some benign cells carry cosines to -0.95.

C Pilot: single-parent evaluation hides parent-locking

A three-parent pilot (2026-08-16) first exposed the confound. The single-parent controller from our original five-condition study transferred lethally to two of three sibling parents (survival 0.00, final Hamming 0.249 and 0.434), while zero-output channel parents beat the K=0 baseline in $3/3$ seeds ($0.029/0.034/0.020$ vs $0.034/0.177/0.028$). The original table's headline gap was therefore part parent-training effect and part parent-seed luck, motivating this study.

D Model and training details

The grid state $x_t \in [0,1]^{96\times96\times16}$ holds 16 channels; the last four are RGBA with alpha last. Perception uses three fixed kernels (identity, Sobel-x, Sobel-y) per channel, giving $p_t \in \mathbb{R}^{48}$:

$$p_t = \big(K_{\mathrm{id}} * x_t,\ K_{\mathrm{S_x}} * x_t,\ K_{\mathrm{S_y}} * x_t\big). \tag{1}$$

The update MLP maps $(48{+}K)$ inputs to 128 hidden units (ReLU) and back to 16 channel increments, with the final layer zero-initialized. Cells update with probability 0.5; a cell is alive when its alpha exceeds 0.1 within its 3×3 max-pooled neighborhood; dead cells are masked to zero.

Table A3: Full 5×5 cross-parent transfer matrix among the five defense parents, both controller replicas (top: e1; bottom: e2). Rows: donor controller; columns: recipient parent. Cells give mean final Hamming / worst-case survival over 3 condition seeds $\times$ 8 held-out damage seeds (24 rollouts per cell). Diagonal (bold): recipient's own controller (survival 1.00 throughout). †lethal (survival 0.00). Across the 40 off-diagonal cells: 23 harmful, 15 indistinguishable from zero-output, 2 beneficial; the beneficial cells are exclusively s4$\rightarrow$s1, the tonic-aligned pair (cosine $+0.99$), in both replicas.

e1 donor	recip. s0	recip. s1	recip. s2	recip. s3	recip. s4
s0	**0.0315**	$0.470/0.00^{\dagger}$	$0.097/0.38$	$0.028/1.00$	$0.052/0.88$
s1	$0.035/1.00$	**0.0306**	$0.023/1.00$	$0.049/1.00$	$0.035/1.00$
s2	$0.029/1.00$	$0.043/0.88$	**0.0201**	$0.033/1.00$	$0.092/0.50$
s3	$0.035/1.00$	$0.036/0.88$	$0.208/0.00^{\dagger}$	**0.0295**	$0.044/1.00$
s4	$0.034/1.00$	$0.030/1.00$	$0.023/1.00$	$0.044/1.00$	**0.0340**
e2 donor					
s0	**0.0291**	$0.108/0.25$	$0.056/0.88$	$0.032/1.00$	$0.039/1.00$
s1	$0.032/1.00$	**0.0292**	$0.025/1.00$	$0.049/1.00$	$0.032/1.00$
s2	$0.041/1.00$	$0.074/0.62$	**0.0219**	$0.033/1.00$	$0.113/0.25$
s3	$0.028/1.00$	$0.058/0.75$	$0.239/0.00^{\dagger}$	**0.0262**	$0.028/1.00$
s4	$0.030/1.00$	$0.030/1.00$	$0.020/1.00$	$0.033/1.00$	**0.0336**

Table A4: Tonic output of the evolved controllers: per-channel mean of m_t, averaged over the two independent evolutions per seed. All ten controllers sit at nonzero, parent-distinct offsets; within-rollout per-channel std of m_t is 0.0002–0.0046 everywhere: flat, tonic output with no lesion-locked response in 10/10 runs.

Seed	mean(m) per channel (avg. of 2 runs)
0	-0.021 $+0.002$ $+0.027$
1	$+0.030$ $+0.012$ -0.026
2	-0.026 -0.032 -0.015
3	$+0.001$ $+0.032$ -0.010
4	$+0.028$ $+0.007$ -0.025

Tonic and phasic modulator dynamics:

$$m_t^{(\text{tonic})} = \alpha\, m_{t-1}^{(\text{tonic})} + (1-\alpha)\, c_t, \qquad m_t^{(\text{phasic})} = m_{t-1}^{(\text{phasic})} \cdot e^{-\Delta t/\tau}, \tag{2}$$

with $\alpha = 0.95$, $\tau = 20$, and the injected level the clipped sum, broadcast to all cells and concatenated to perception (input dim $48 + 3 = 51$).

Modulator inputs during parent training. The `mod_noise` flag was false in every parent-training run: the injected level c_t was pinned to the neutral vector $m{=}0$ for all 8,000 steps (the flag exists to train under random modulator inputs and exercise the channel weights; it was never enabled). Two consequences follow. First, the channel-input weights of the update MLP receive zero gradient whenever the injected level is zero, so they remain at their initialization values in every trained parent; the $K{=}3$ parent evaluated at $m{=}0$ is therefore functionally a $K{=}0$ network with a different parameter-initialization draw (the extra first-layer columns shift the RNG stream). Second, "channel-aware training" as run isolates *training with the architecture present*, not modulatory experience: any effect must act through initialization or through training with the enlarged input layer. At $n{=}5$ the preregistered 5/5 outcome is a one-sided sign test at $p{\approx}0.031$ (the $\geq 4/5$ bar corresponds to $p{\approx}0.19$ under a symmetric null) — consistent with a real effect, but not separable from initialization luck, and further entangled with the shared-state asymmetry of §3. Retraining parents with random modulator inputs and adding $K{=}0$/$K{=}3$ initialization pairs have since been run under preregistration (Appendix J): exercised-channel parents gain a genuine corrected-protocol advantage ($5/5$), while fresh silent pairs show none — the training effect is real exactly when the channels are exercised. Unaffected by this confound: the evaluation-time probes, since constant levels alone swing mean Hamming from 0.003 to 0.93 through the same (randomly initialized) channel weights, the pathway the controllers and transplants act on.

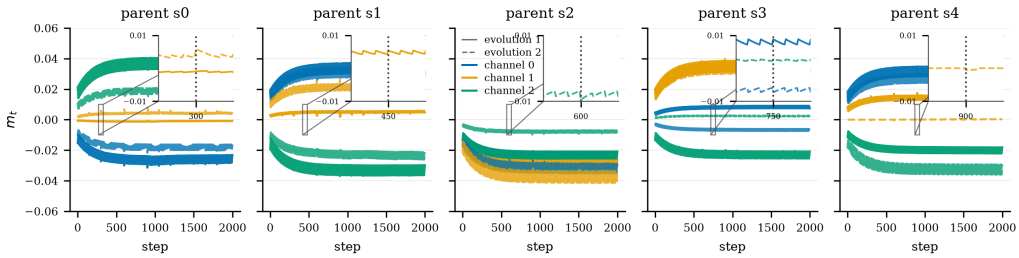


Figure A1: Controller output m_t over a full evaluation rollout, per parent seed, both independent evolutions overlaid (solid/dashed), one line per modulator channel. After a brief initial transient while the perceptual state settles, every controller holds a constant, parent-distinct offset through all lesion events: tonic calibration, not an event-locked policy. Note that the two independent evolutions of the same parent sometimes settle at slightly different setpoints.

Parents train on a 40×40 emoji target zero-padded to the canvas, with pool-based training (pool 1024, batch 8, worst-in-batch reseeding), random square-erasure damage mid-episode, MSE loss on RGBA, and Adam at 10^{-3} for 8000 steps. A first attempt at 2×10^{-3} diverged to NaN near step 5532 via a late-stage loss-spike cascade; 10^{-3} absorbs the identical spike.

E Perception-radius sweep

A single-lesion sweep (radii $\{2, 4, 8, 16\}$, single- and multi-site, with and without modulator channels) maps the boundary of local repair. Recovery is near-perfect for small lesions and degrades once lesions exceed the effective perception radius. Two failure signatures motivate this work: severed fragments that survive but cannot reattach, die, or decide what to grow into, and persistent half-alpha scar tissue at wound sites. Both are information failures: the misbehaving cells are exactly the cells cut off from global context.

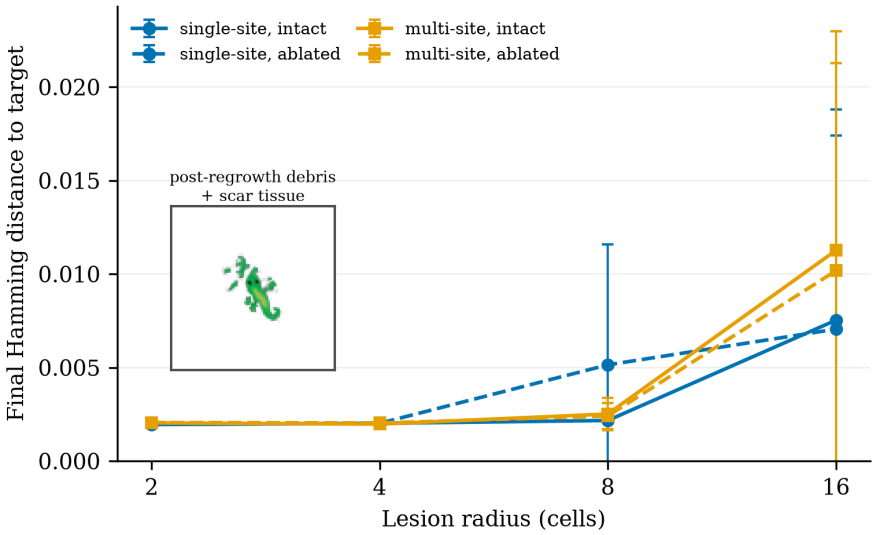


Figure A2: E1 lesion sweep. Final Hamming distance as a function of lesion radius and kind, with and without modulator channels (mean $\pm$ SD over 5 damage seeds). Inset: post-regrowth debris and scar tissue.

F Calibration probes

Collapse probe. The target mask is only 4.6% alive, so a fully-dead grid scores Hamming ≈ 0.046, close to a struggling organism's ≈ 0.02. We verified empirically that the all-dead state scores 0.0461 against struggling-neutral's 0.0205: death loses by $2.2\times$, so the alive-fraction floor is disabled (0.0). Notably, the runbook-default floor of 0.3 would have *penalized every healthy candidate* (a correct lizard lives at ≈ 0.057 alive), pushing evolution toward overgrowth, a case where calibrating from the experiment's own data overrode a plausible default.

Initial step-size probe. Perturbing the neutral controller by Gaussian noise at increasing σ (25 samples each, hard-regime evaluation) showed: $\sigma \leq 0.01$ stays healthy (some samples beat neutral); $\sigma = 0.05$ is bimodal; $\sigma = 0.3$ lands *every* sample in saturated overgrowth (alive $\to 0.96$). A first evolution run with the library-default $\sigma_0 = 0.3$ froze at fitness 0.150 from generation 2 onward, $7\times$ worse than doing nothing, because CMA-ES was ranking overgrowth-degree noise. $\sigma_0 = 0.01$ brackets the neutral point and the same budget converged smoothly (Figure A3).

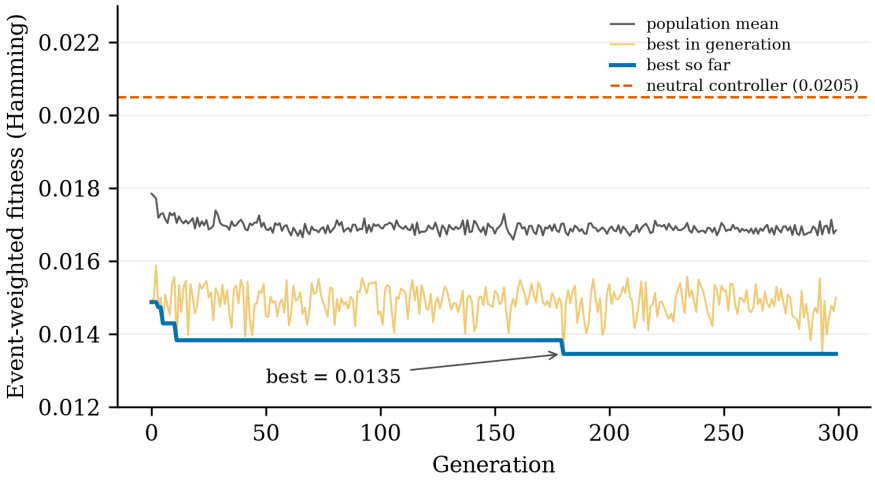


Figure A3: CMA-ES fitness over 300 generations (σ_0=0.01). Dashed line: neutral controller (0.0205). Best evolved fitness: 0.0135, a 34% improvement with no stall.

G Evolution objective

We evolve the controller with CMA-ES [13] (Evosax implementation [14]), population 64, initial step size σ_0=0.01, 300 generations, neutral (all-zero) controller as the initial mean. Fitness is the event-weighted mean Hamming distance to the target alpha mask over full rollouts on the eight train damage seeds:

$$f(\theta) = \frac{1}{\sum_t w_t} \sum_{t=1}^{T} w_t \, H_t(\theta), \qquad w_t = \exp\Big(- \frac{t - \tau_{\text{last}}(t)}{\tau_w} \Big), \tag{3}$$

where $\tau_{\text{last}}(t)$ is the most recent lesion step and $\tau_w = 150/3 = 50$. The recency kernel resets at each lesion, so slow repair is expensive even when endpoint Hamming is identical, so it de-saturates the metric and prices repair speed directly. Fitness is not the reported metric: all results report unweighted trajectory statistics on held-out seeds. Two calibration controls make this landscape interpretable: a collapse probe (verifying the objective cannot be gamed by alpha suppression) and an initial step-size ablation (showing σ_0=0.3 saturates the first population in overgrowth where no repair signal exists), both detailed in Appendix F.

H Single-parent five-condition study

Five conditions share the same protocol, horizon, and damage seeds: **closed-loop** (evolved controller); **static** (the evolved controller evaluated once at $t{=}0$ and held constant, isolating temporal modulation from the evolved tonic level itself); **constant** (fixed tonic, grid-searched over $\{-1, -0.5, 0, 0.5, 1\}$ on train seeds); **random** (uniform $[-1, 1]$ each step, actuation-matched noise); and **no modulation** ($K{=}0$ parent). All experiments ran on a single rented A100; parent training took ≈20 min/model, evolution ≈2.5 h, total project cost under $15.

Modulation helps substantially (RQ1). Relative to no modulation, modulated conditions recover $2.2\times$ more completely (final Hamming 0.028–0.030 vs. 0.063) and sustain $\approx 4\times$ less cumulative damage (AUC 0.016–0.017 vs. 0.064). The wounds exceed the perception radius, so this improvement is attributable to the broadcast channel, the only non-local pathway, which repairs damage that local rules cannot diagnose.

A regime boundary, not a failed ablation (RQ2). The three modulated conditions are statistically indistinguishable: final Hamming spans 0.028–0.030, half-life 6.4–7.2 steps, all differences within noise (pairwise effect sizes < 0.15). We read this as a measurement of the *regime*, not a failure of the controller: the damage process is stationary and memoryless (events are i.i.d. in size, number, and position), so the optimal release policy is time-invariant and a fixed tonic gain is structurally adequate. There is no temporal structure for adaptive scheduling to exploit. Evolution's measured contribution is automatic discovery of the near-optimal tonic level in 2.5 hours from a neutral start, without the hand search the constant condition required.

Random modulation is lethal. Random actuation kills every run (survival 0.00, final Hamming 0.825): pilot probes show constant levels alone swing mean Hamming from 0.003 to 0.93. The modulator channels are a high-gain control pathway, not a free architectural bonus; the identity of the release schedule determines whether the organism lives.

Metric artifacts of long-horizon evaluation (RQ3). Two artifacts surfaced only under repeated damage. First, repair half-life is defined per lesion relative to each post-lesion jump; on the baseline's chronically damaged morphology, new lesions often land on already-degraded regions, producing no measurable jump and scoring as zero half-life, so the baseline's 2.7 steps does *not* indicate fast repair. Second, residual Hamming in all conditions is partly attributable to locomotion drift during regeneration rather than permanent damage; the metric conflates spatial translation with morphological error. Both are documented so that future benchmark users can avoid them; the camera-ready revision reports a translation-aligned metric and an initial-state-provenance control for the main comparison (Appendix I).

I Post-hoc controls: initial-state provenance and translation-aligned scoring

Added for the camera-ready revision at reviewer request. Two protocol properties could inflate the preregistered channel-training comparison: the shared $t{=}0$ state is grown by the $K{=}3$ parent, placing the $K{=}0$ condition in an out-of-distribution initial state (§3); and fixed-frame final Hamming conflates shape error with whole-organism drift (Appendix H). The controls rerun the exact evaluation protocol (held-out damage seeds 10000–10007, five condition seeds, $T{=}2000$, identical key streams and the first controller replica's config wiring) with two changes, one at a time: **k0_own** grows the $t{=}0$ state with the $K{=}0$ parent itself (the shared state is already the $K{=}3$ parent's own-grown state, so the provenance comparison is symmetric), and **aligned** replaces final Hamming with the minimum mismatch over integer translations of up to ± 10 pixels. Plain columns reproduce the preregistered numbers (s1: 0.161 here vs. 0.163–0.175 in Table 1).

Three readings. (i) The shared-state protocol itself flips one seed: on plain Hamming from own-grown states, s4 reverses (0.029 vs. 0.033), breaking the $5/5$. (ii) The fragile s1 pair is hit hardest: from its own state, final Hamming halves ($0.161 \rightarrow 0.088$) and survival rises from 0.10 to 0.50 (0.675 aligned) — yet a clear gap remains (0.026 vs. 0.071 aligned), so the fragility is substantially intrinsic to that initialization draw, amplified but not created by the shared state. (iii) Aligned scoring dissolves the healthy-parent gaps: all four non-s1 comparisons sit within noise

Table A5: Post-hoc controls (mean over 40 runs per cell). Survival in parentheses where below 1.00; for the s1 own-grown row the bracketed value is survival under the aligned metric.

	K=3, m=0 (own-grown)		K=0, shared state		K=0, own-grown	
Parent	plain	aligned	plain	aligned	plain	aligned
s0	0.029	0.021	0.035	0.018	0.037	0.019
s1	0.038	0.026	0.161 (0.10)	0.144	0.088 (0.50)	0.071 [0.675]
s2	0.018	0.011	0.030	0.013	0.028	0.010
s3	0.028	0.018	0.049	0.018	0.033	0.013
s4	0.033	0.020	0.043	0.026	0.029	0.019

(largest $|\Delta| = 0.006$ against SDs of 0.01–0.02) and nominally favor K=0. The preregistered $5/5$ therefore decomposes into protocol-provenance and drift effects on healthy parents plus one genuinely fragile parent. What these controls do *not* support is a channel-experience mechanism: the channels were silent during training (Appendix D), and the surviving s1 effect is a property of the parent draw. Separating initialization luck from any residual architecture effect requires the added initialization pairs of the planned follow-up (§5). Script: `src/posthoc_metrics.py`; data: `experiment_results/20261011_posthoc_controls/`.

J Preregistered resolution: channel-exercise retrains and init-luck nulls

Added at revision, preregistered before any run (decision rules fixed in `experiment_results/20261012_p1_p1b/PREREGISTRATION.md`). Both arms retrain fresh parent pairs per seed with the study recipe (Adam 10^{-3}, 8000 steps, all else per Appendix D) and evaluate with the Appendix I battery. **P1** (seeds 0–4) trains each pair's K=3 parent with *random* modulator inputs during training ($c_t \sim U[-1,1]^3$ drawn per pre-/post-damage phase — the intended use of the `mod_noise` flag, which was off for the study parents). **P1b** (seeds 10–13) retrains the original *silent*-channel condition on fresh draws, measuring the init-luck null of the paired gap.

Table A6: P1/P1b fresh pairs under the Appendix I battery (mean over 40 runs; survival in parentheses where below 1.00). $g^{\text{ali}}_{\text{own}}$ = aligned, own-grown gap, positive favoring K=3; $g^{\text{pln}}_{\text{shr}}$ = the same pair under the original protocol (plain, shared state).

Pair	K=3 own (pln/ali)	K=0 shared (pln/ali)	K=0 own (pln/ali)	$g^{\text{ali}}_{\text{own}}$	$g^{\text{pln}}_{\text{shr}}$
P1 s0	0.026/0.010	0.071/0.054 (0.88)	0.025/0.018	+0.008	+0.045
P1 s1	0.030/0.016	0.046/0.046	0.046/0.046	+0.030	+0.016
P1 s2	0.026/0.009	0.125/0.107 (0.17)	0.045/0.032 (0.95)	+0.023	+0.099
P1 s3	0.021/0.010	0.051/0.031	0.034/0.022	+0.012	+0.031
P1 s4	0.035/0.017	0.062/0.044	0.030/0.020	+0.003	+0.026
P1b t10	0.028/0.014	0.059/0.026 (0.97)	0.029/0.012	−0.002	+0.031
P1b t11	0.030/0.012	0.068/0.040 (0.97)	0.033/0.028	+0.015	+0.037
P1b t12	0.035/0.019 (0.90)	0.114/0.098 (0.33)	0.022/0.014	−0.005	+0.079
P1b t13	0.029/0.010	0.024/0.015	0.020/0.012	+0.002	−0.005

P1: channel exercise restores the effect. Exercised-channel parents beat their paired K=0 siblings in $5/5$ seeds under the corrected protocol (median $g^{\text{ali}}_{\text{own}} = +0.012 > 0.005$; the preregistered rule is met).

P1b: silent channels alone do not. The fresh silent pairs show no corrected-protocol effect ($2/4$ positive, $|\Delta| \leq 0.015$) while the *original* protocol scores $3/4$ of the same pairs as large positives (up to $+0.079$) — a controlled reproduction of the manufacturing process behind the study's 5/5. The shared-state penalty on K=0 replicates across the new pairs (e.g. P1 s2's K=0 parent: survival 0.95 from its own state, 0.17 from the shared state).

Reading. Because P1 modulates with *random* inputs, the restored effect is developmental channel exercise, not control: what training needs is signal in the channels, not a policy choosing it — the mirror image of the run-time finding that evolved policies reduce to tonics. Together with Appendix I, the attribution reads: silent architecture alone, no effect (the original $5/5$ was proto-

col artifact); exercised channels, robustness; closed-loop control over exercised channels, nothing beyond a setpoint.

K Reproducibility

Parents train in ≈20–30 min each on one A100; the defense study ran two independent 300-generation controller evolutions per parent (10 total), each ≈2.5–3 h on the study A100, for a total defense-study cost of ≈$25 of rented GPU time. The transfer matrix is evaluation-only over existing artifacts: 40 off-diagonal evaluations, ≈2 minutes of A100 time per full 5×5 replica matrix. Preregistration, analysis script, per-seed artifacts (controllers, parents, trajectories, m_t series), and the mechanical outcome output are in the repository, timestamped before the results. The ten parent checkpoints (`k0_s0..4`, `k3_s0..4`), the data behind every figure (including the E1 lesion sweep and the transfer and tonic-transplant matrices), and the camera-ready control script (`src/posthoc_metrics.py`; data in `experiment_results/20261011_posthoc_controls/`) are in the same repository, and `paper_drafts/make_figures.py` maps each figure to its data file. The preregistered follow-up (P1/P1b: 18 fresh parent checkpoints, per-pair CSVs, mechanical analysis) is under `experiment_results/20261012_p1_p1b/`. Code is available at https://github.com/jverene/neuromodulation.